

MAST32001 Week 4 Summary

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This week, we went past the basic bootstrap. We used the bootstrap for two new jobs. One was hypothesis testing and the other was fitting regression. We also started cross-validation.

The main idea for a bootstrap test is simple. We must act as if the null hypothesis is true. We do this when we sample. We can do this in two ways. First, we can fit a model under the null. Then, we simulate new data from that mode. Second, we can shift the sample. We make its mean match the null value. Then we resample. We also compared this with permutation tests. Permutation tests work because data have symmetry. They do not need us to plug in estimates.

For regression, we learned two main bootstrap methods. The residual bootstrap keeps the x value fixed. It only shuffle the errors. This method makes sense when we chose the x values ourselves. It also makes sense when all errors have the same spread. The pairs bootstrap shuffles whole (x, y) pairs together. This method is better when the x values are random. Sometimes the errors do not have the same spread. This problem is called heteroscedasticity. Then the residual bootstrap does not work well. We learned about the wild bootstrap for this case. The wild bootstrap keeps each error tied to its own x . It just flips the sign of the error. Then we looked at time series. The usual bootstrap breaks time series. It mixes up the order. The block bootstrap fixes this problem. It resamples whole chunks instead of single points.

The second lecture was about prediction. It asked how well a model predicts data it has not seen. The main warning is that training error lies. If we score a model on the same data we trained it on, it looks better than it really is. We split prediction error into three parts. The first part is noise. We cannot get rid of noise. The second part is bias. Bias happens when the model is too simple. Third part is variance. Variance happens when the model jumps around a lot if the data change. When a model gets more flexible, bias goes down. But variance goes up. So the test error makes a U shape.

Then we learned cross-validation. We split the data into folds. We train on most the data. We test on the held-out fold. We rotate the folds. Then every point gets tested one. Leave-one-out is the extreme version. Linear regression has a shortcut. We do not have to refit the model n times.

The tricky part was model selection. Picking the “best” model with CV adds its own kind of overfitting. If we try 100 models, the winner partly got lucky. There are fixes. One fix is nested cross-validation. Another fix is to make sure test data never leaks into preprocessing. Another fix is use AIC or BIC. These methods charge extra for each parameter we dd. We also saw that the CV setup must match the problem. For time series, we can only predict forward. For data that comes in groups, we hold out whole groups.

The easiest thing this week was writing the basic residual and pair bootstrap in R. It is a short recipe. The plots are wasy to read. The hardest thing was understanding why the smallest CV score is not honest by itself. It was also hard to understand why that problem needs cross-validation.